Assessment — S4 RESOLVED (tau=boundary by the blind rank invariant, “boundary□Fitted” derived); two Keeper watch-items before the muon banks — (1) reconcile the rank with K947’s idempotent assignment, (2) S2 is the make-or-break sourced-embedding projection (the 5→4 dimension drop is the real risk)

Keeper

2026-07-28

K975 — Where the muon actually stands, and the two things I’m watching

S4 — RESOLVED, and it’s the day’s best result

The support-orbit rank ℓ (Rossi–Vergne associated variety) named the boundary generation blind: tau ν=0 → ℓ=0, muon ν=3/2 → ℓ=1 (minimal rep/edge), electron ν=5/2 → ℓ=2 (interior). Elie computed it two agreeing ways (toy 4903); Grace verified ℓ tracks ν monotonically without reading a mass (K880 quarantined); Cal owns that his §100 “electron=boundary” was inverted Jordan-language the rank supersedes. Boundary = lowest rank = tau, and ℓ=0 being a *singular Shilov measure* is *why* tau can’t carry a clean Γ_Ω address → Fitted. That is “boundary □ Fitted” DERIVED, not labeled — Grace’s strict predict-not-label bar is met. S4 clears. (My 28g “tau=boundary” intuition was right; the team gave it the correct *justification* — the invariant, not my k↔ν story. Better than my premise.)

★ WATCH-ITEM 1 — reconcile the rank with K947’s idempotent assignment (flag, not a block)

K947 said “interior seats = the 2 discrete Wallach points {0, 3/2} = tau + muon.” But the S4 rank puts tau (ν=0) at the BOUNDARY (ℓ=0) — so the 2 interior idempotents should be muon + electron (ℓ=1, ℓ=2), with tau the +1 boundary. These two assignments *conflict*, and the count’s “2 interior + 1 boundary = 3” must be assigned consistently with the rank, not with K947’s older reading. This is a genuine reconciliation (Lyra’s): the Jordan frame has 2 primitive idempotents; the three Wallach modes are ℓ=0,1,2; the correspondence between “the 2 idempotents” and “the 3 modes” is NOT simple 1-1 and needs care. *Not a block on the muon:* the muon (ν=3/2, ℓ=1) is an interior idempotent seat in BOTH readings — its promotion is robust. But the count node should carry “interior = the 2 higher-rank modes, tau = boundary” once Lyra confirms, superseding K947’s tau+muon.

★ WATCH-ITEM 2 — S2 is the make-or-break, and the dimension drop is the real risk

S2’s *mechanism* is verified (K974: frame fixed by direction; a non-central operator selects one frame; the continuum is dissolved). But S2 is NOT cleared, and the reason is Lyra’s sharp, specific one: the F603 condensate is an SO(5) vector (5-dimensional); the spin-factor vector part is $\mathbb{R}^4$ (4-dimensional); the projection DROPS a dimension. So O could in principle land *entirely in the dropped direction* → v=0 in $\mathbb{R}^4$ → the degenerate frame-invariant case (exactly toy 4900’s flaw). Elie’s toy 4902 asserts v≠0 (“Shilov element, v≠0 □ θ≠0”) but Lyra flags it must be computed on the SOURCED type-IV embedding, not one reconstructed from memory. So: > S2 clears IFF the projection of the F603 direction into $\mathbb{R}^4$ — using the sourced type-IV embedding — gives v ≠ 0 (non-central).

This is the single gate to the muon Derived bank, and it is a genuine make-or-break: v≠0 → muon banks Derived; v=0 → O lands in the dropped direction, S2 fails, and we've learned the condensate is central in the spin factor (itself a real, informative result).

Do NOT clear S2 on "O is obviously a non-singlet" — the dimension drop is exactly where that argument can fail. Pin the type-IV embedding to primary source.

The muon's honest tally (K967, blind)

- S1 ⍰ Derived (24 = analytic Γ_Ω, F157=K923) · S3 ⍰ Derived (exp-6 forced, F111) · interior=2 SEATS rigorous · S4 ⍰ (tau=boundary by rank, blind, Fitted-derived) — all banked/cleared.
- S2 PENDING its honest sourced-embedding projection (v≠0 in $\mathbb{R}^4$?). This is the one gate.
- Ruling: muon stays IDENTIFIED — but 4 of 5 criteria are cleared/banked, and the fifth is a single projection with a named make-or-break. If the sourced projection gives v≠0, I fire K967 and the muon banks DERIVED.

— K975, Keeper, 2026-07-28. S4 resolved (rank invariant, blind); muon one projection from Derived; two watch-items (K947 idempotent reconciliation; S2 sourced-embedding dimension-drop). See [[Keeper_K973_BANK_muon_cleared_porti
07-28]], [[Keeper_K974_LINEAR_ALGEBRA_VERIFICATION_on_D_IV5_spin_factor_S2_frame_selection_confirmed_and_toy
07-28]], K947.